

Lucas Lopatka

Senior Full-Stack & Platform Engineer

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Born May 1986 / 40yo · Canadian–Thai dual citizen · English (Fluent) · French (Fluent) · Thai (Fluent)

Full-stack & platform engineer with 15+ years shipping production software across payments, identity, government and e-commerce. I work from fundamentals over frameworks and move fluidly between stacks.

Languages

Top: HTML/CSS (20 years) · JavaScript (20 yrs) · TypeScript (10 yrs) · PHP (15 yrs) · Python (3 yrs) · Go (3 yrs)

Other: C# (3 yrs) · Java (2 yrs) · Ruby (2 yrs)

Backend & APIs NestJS (5 yrs) · Node/Express (10 yrs) · Laravel (10 yrs) · Flask (3 yrs) · REST / OpenAPI (15 yrs) · gRPC (3yrs) · GraphQL (1yr) · WebSockets (15 yrs) · SOAP / XML-RPC (3yrs legacy), Spring Boot (3yrs) · Quarkus (Keycloak) (2 yrs)

Frontend React · React Native / Expo (8 yrs) · Vue.js (8 yrs) · Angular / AngularJS (1 yr)

Data PostgreSQL / PostGIS (10 yrs) · MySQL (10 yrs) · SQL Server (SQL) (3 yrs) · MongoDB (5 yrs) · Solr (2 yrs)

Data modeling with ORM eg. Prisma · TypeORM · EF · Doctrine (15 years)

DevOps CI/CD (10 yrs) · Docker (7 yrs) · Kubernetes (3 yrs) · OpenShift (4 years) · Helm (5 yrs) · FluxCD (3 yrs) · Istio (6 months)

Cloud AWS Cloud Stack (7 years) · Azure Cloud Stack (3 years)

Messaging RabbitMQ (4 yrs) · NATS (CloudEvents) (2 yrs) · Kafka (2 yrs)

Identity & security Keycloak / Red Hat SSO (7 yrs) · Microsoft Entra + Graph (6 months)

Authentication for APIs using JWT and/or OAuth2 / OIDC (PKCE) (10+ yrs) · PCI-DSS scope (3 yrs)

AI Anthropic Claude / Claude Code (1 yr) · MCP servers (Node + Laravel) for AI-agent tooling (1 yr)

Experience

Contract + Temporary Software Developer — Skin Tyee First Nation

May 2026 – July 2026 · Southbank, BC – Temporary role, 3-month contract, NDA does not allow me to discuss further details

Following a disaster incident, I was contracted to scaffold and set up a Canadian First Nation's band's technology end-to-end — hardware setup, domain architecture, set up roles and access using Entra ID for SSO, Azure cloud, network, data sovereignty and encryption solutions design, security and disaster recovery policy and testing. Acted as temporary domain architect.

- Used AI to build a role-gated React Native community app (dashboard, directory, events, transparency, billable time-keeping) and a Consultation Referrals system with PostGIS maps and SharePoint + Azure Blob storage, all deployed as Azure Container Apps to Azure Cloud via Azure DevOps.
- Used AI to design and deploy a contract-first NestJS API (OpenAPI, PostGIS). Advisory on ERP integration (Sage Intacct) and an e-sign + documents portal, all deployed as Azure Container Apps to Azure Cloud via Azure DevOps.

Technical Lead — CrashPad411

Feb 2024 – Jan 2026 · Full-time · Remote (USA) · crashpad411.com

- Technical Lead for CrashPad411.com — the #1 rental listings site for air-crew worldwide. Owned the architecture and led all six repositories (authoring ~1,400+ of ~2,600+ commits), modernizing the full stack legacy applications.
- Rebuilt users, payment and subscriptions engines according to SOLID principles, with an idempotency-aware, events-driven architecture leveraging webhooks and a full audit log, as opposed to the legacy timeout + retry at the service layer, resulting in an auditable, Stripe-reconcilable ledger and the elimination of the previously ubiquitous duplicate payment issues plaguing the platform.
- Implemented additional solutions for Stripe Connect payouts, custom in-house subscriptions, and AutoPay automatic rent payments from tenants to landlords.
- Led the Vue 2 → Vue 3 (SSR on Express) migration and PHP 7→8 / Laravel 8→11 upgrades. Leveraged AI tooling / Claude Max in conjunction with various specifically designed tests in Cypress to deliver this change within months instead of a year.
- Shipped iOS + Android apps without the need for additional native apps by migrating from multiple React-Native apps to one.

- Consolidated multiple full-API applications for white-label customer / pre SaaS + VM hosted instances, into a single SaaS monolith, reducing unnecessary or duplicated code and logic across the organization by over 70%, and saving 60% overhead on cloud-costs and hosting.
- Drove a full UI/UX redesign of the Vue frontend application, and was personally responsible for everything from mockups and wireframes, to manually coding the entire new frontend application for the web. I also drove a corresponding full UI/UX redesign of the admin management portal, and led a frontend team of 5 part-time and 1 full-time developer in its re-implementation.
- On the backend I maintained a large Laravel 11 REST API (600+ routes) documented with Swagger → OpenAPI 3 — JWT auth, Stripe Connect + Stripe webhooks, AWS S3 uploads, real-time WebSocket + email + push notifications, Laravel Nova admin and Horizon queues.
- I re-architected the 9-node white-label network of custom users to the new platform, featuring a new self-service onboarding portal (Python/Bottle + Vue 3 + MongoDB);
- Introduced CI/CD — an organizational first — and authored Gitlab CI/CD pipelines deploying to DigitalOcean, later migrating to Azure DevOps CI/CD with self-hosted Docker agents to save on build costs.
- Introduced AI-enhancement to the platform, using an Laravel MCP server exposing the listings API to AI — we used an in-product chatbot, MCP-driven content generation, listing recommendations and user-settings configuration to help customers use the product more easily, reducing technical support frequency and timelines.

Senior Software Engineer — SSO / identity (Java · Keycloak · Quarkus) — Opn / Omise

Apr 2021 – Mar 2024 · Full-time · Bangkok, Thailand · omise.co

- Technical lead for Opn/Omise's single sign-on unifying dozens of previously siloed products — Merchant Dashboard v1 (Rails) + v2 (React), the UMay+ Virtual Credit Card app, the Payment Links+ suite (frontend / admin / system-admin), an app store and Xero, among other programs — at a US\$19B+ payment gateway (20,000+ merchants; public case studies include BMW, McDonald's Thailand, King Power, Allianz Ayudhya, Minor, Pomelo and The Pizza Company), with zero-breakage backward compatibility for live merchant API tokens. I worked directly with multiple Technical Product Managers, Engineering Managers, Team Leads and at times the SVP of Engineering to achieve organizational goals.
- Not being a Java developer, but having encountered Keycloak previously at the BC Government with CGI, I knew we could solve our SSO requirement with an open-source platform, without expensive enterprise solutions and per-user licenses, by implementing custom extensions across several Keycloak SPIs — including a completely custom User Storage Provider (User Storage SPI) federating the existing user store. Other customizations included custom OAuth scope / claim mappers (Protocol Mapper SPI); custom-branded login / account / email themes (Theme SPI, FreeMarker), and service reporting / monitoring using DataDog and custom DataDog dashboards. I championed this initiative — despite initial pushback from concerned parties — across multiple organizational divisions and Agile teams, and was the principal developer of the corresponding solution, and Keycloak continues to be used as the SSO solution organizationally as of today.
- As part of implementing org-wide SSO, I solved the issue of bridging legacy Rails sessions + live API tokens — having never worked with Ruby on Rails - into the OIDC model via a gRPC + protobuf service the Keycloak provider called on each login (surfacing the token as a custom OAuth scope); solved auth-load at production scale with a bespoke, RAID-like Infinispan distributed cache and auto-scaling Kubernetes replica-sets of nodes — load-tested with JMeter to ~300 logins/sec/node, etc. I worked with the elite Omise Core team on this to achieve mutual goals with zero production downtime.
- I carried our Keycloak migrations through its WildFly → Quarkus distribution migration — the new build / config model (kc.sh build), provider packaging, and dev-mode live reload; ran and debugged both distributions locally. Successfully migrated, incrementally, from Keycloak 15-23 during my time at Omise with zero *production* downtime. Furthermore, I self-managed internal support tickets and requests for feature from our internal team of users, as SSO impacted both internal and external users. This was not always without scheduled downtime.
- As a Sr. Software Engineer in a hub-and-spoke GitOps environment, I was responsible for building, maintaining, and owning all CI/CD pipelines for all applications I worked on, and also owned the corresponding Kubernetes + Helm + FluxCD (GitOps) configurations, deploying using an Istio service mesh using a primarily blue-green strategy across A/B failover domains (Buildkite).
- I led frontend product development for the UMay+ Virtual Credit Card application — I architected and built the entire frontend end-to-end and contributed to backend Go service development — a PCI-scope, gRPC-powered suite of backend micro-services in Go (delivered with our corresponding backend team) linking a merchant's profile to the UMay+ issuing backend, with full KYC (identity verification) and OTP-based setup / verification. Live: vcard.issuing.opn.ooo.
- In addition to my lead of the SSO project, I was also a Sr. Engineer delivering functionality and bugfixes for the Payment Links+ & Merchant Dashboard v2 suite of applications (React, Next.js, GraphQL, Node, Express) — representing several of the dozens of

products the SSO unified: Payment Links+ (frontend / admin / system-admin) generates customized payment links, invoices and receipts to drop into websites, chats, emails and social DMs; Dashboard v2 was a Stripe-like merchant dashboard for managing customers, receipts and transactions. Live: omise.co/en/payment-links-plus.

- Reliability at payments scale — Omise publicly cites 99.99% availability and ~100M transactions/day (omise.co); the SSO sat on the critical login path across A/B blue-green failover domains.
- Impact: eliminated the need to implement authentication and authorization for all Omise merchant or customer facing and internal apps, ~60% fewer login-support tickets per application, tons of money saved using the open-source Keycloak vs the expensive Red Hat SSO enterprise licenses — and still in production, live on various apps at omise.co. Unfortunately, my NDA does not allow me to disclose any concrete supporting business financials or metrics achieved.

Senior Full-Stack Engineer — Frontend Apps Lead — CGI — BC Provincial Government

Nov 2019 – Apr 2021 · Full-time · Victoria, BC · names.bcregistry.gov.bc.ca · fom.nrs.gov.bc.ca

- I was employed in a role similar to a forward deployed engineer, where I would work on-site, with government ministries and their officials and their teams, designing and implementing custom solutions for various organizations, departments and ministries of the British Columbia provincial government including Service BC, BC Registries, the Ministry of Justice, and the Ministry of Forests. I worked directly with government-side project managers, product owners, and technical delivery teams in CGI (solution architects) to deliver on-site, fully scoped project plans and proposals, and worked on ensuing deliverables as a Senior Full-Stack Engineer.
- Open-source digital government for British Columbia on its OpenShift platform — ~1,593 commits / 179 PRs across 6 bcgov repos as bcdevlucas (github.com/bcgov/namex · github.com/bcgov/jag-shuber-frontend · github.com/bcgov/nr-fom).
- Projects I worked on include but are not limited to:
 - Name Request / NameX (Service BC · BC Registries): the business-name examination + approval engine and public UI — synonyms API + Python client, word classification, extra-provincial (XPRO) handling, name-conflict analysis, credit-card payments and NATS event queues (Python / Flask / Marshmallow, Postgres, Solr, Keycloak; Vue UI).
 - Sheriff scheduling (Justice & Attorney General): roles/auth, admin, scheduling and UTC cross-timezone handling with Liquibase (React + API).
 - Forest Operations Map (nr-fom). Hands-on with Keycloak SSO integrations and OpenShift deploys.
- Live: names.bcregistry.gov.bc.ca · fom.nrs.gov.bc.ca. Recommended on LinkedIn by Lorna Trent and Arturo Reyes Lopez (“I highly recommend Lucas for Software Architect positions”).

Senior SWE / Frontend Apps Lead — Agereh Technologies Inc.

Jul 2017 – Feb 2019 · Full-time · Edmonton, AB · agereh.com

- An AI and intelligent infrastructure company delivering operational intelligence for transportation, logistics, and airports. Known as Carbeeza Inc. before rebranding in September 2025. They are listed on the Toronto Stock Exchange as TSEX:AUTO.
- I led the frontend team, and owned the design and development of the original consumer portal and dealership / finance hybrid applications (TypeScript / React) through the startup’s first funding round. I worked with UX/UI designers and Jr. Software Engineers to accomplish frontend goals in a Scrum / Agile working environment.
- Built and tested various Node.js / Express micro-services connected via RabbitMQ message brokers.
- Containerized and deployed front-end applications using Docker and OpenShift (Kubernetes-based). Optimized container images using small base images for faster pulls and autoscaling, higher pod density, and smaller attack surfaces.
- Worked against the core Java / Spring Boot + Swagger REST monolith through documented service contracts — the contract-first Swagger → OpenAPI 3 approach I have used at every role since.

Solutions Developer — Caffé Tech

Jul 2015 – Jul 2017 · Full-time · Edmonton, AB · caffetech.com

- Rebuilt an espresso-machine / parts vendor’s organization using a custom OpenCart installation connected to a network of service APIs delivering complete Intuit QuickBooks Online integration.
- Built a custom POS and QR based menu ordering system.
- Built custom Canada Post + Purolator shipping (thermal labels) add-ons for the website.
- Catalogued a 5,000–10,000-part inventory into QuickBooks, Chart of Accounts rebuild, etc.
- Managed Microsoft domain for employees using Microsoft SBS + Windows 10 PCs + Parallels for Mac.

Software Engineer (Level 3) — Alberta Motor Association

Feb 2013 – Dec 2014 · Full-time · Edmonton, AB · ama.ab.ca

- Cloned the legacy FRONT-TIER insurance ERP into an in-house web platform (AgentView) reaching every AMA policyholder in Alberta, on a 6–7-person team.
- Built a custom Telerik-Kendo JavaScript rendering framework for the massive vehicle-insurance forms (~30–50% faster agent claim calls) and worked on integrations to CGI AutoPlus, Kanetix and Equifax (largely SOAP).
- AgentView was internal software used by AMA insurance agents across Alberta (no public URL) — my focus was the forms-rendering framework and the third-party rating/credit integrations.

Full-Stack Engineer — Carsgone.com

May 2011 – Oct 2012 · Full-time · Edmonton, AB · carsgone.com

- Architected an original virtual-dealership web app + API (PHP / Kohana, Sphinx search, a multi-threaded feed parser) with Strathcom Media, a local partner.
- Built for a used-car marketplace syndicating dealer inventory — the multi-threaded parser ingested and normalized large nightly dealer feeds into Sphinx for fast search.
- Recommended by Ray Robichaud, President: “Lucas has done great work with us... a competent and valuable web developer.”

Selected live work & open source

- Live systems: crashpad411.com · omise.co/en/payment-links-plus · vcard.issuing.opn.ooo · linksplus.omise.co · names.bcregistry.gov.bc.ca · fom.nrs.gov.bc.ca · caffetech.com
- Open source: ~1,593 commits / 179 PRs across 6 BC-government repos as github.com/bcdevlucas — e.g. github.com/bcgov/jag-shuber-frontend (847), github.com/bcgov/jag-shuber-api (380), github.com/bcgov/namex (226) — plus 50+ public repos at github.com/bluecollardev (my QuickCommerce platform in PHP · C# · Python; Mediashare in TypeScript).

Side projects - Founder / Principal Web Developer - Blue Collar Development

2007–2026 · Thailand / Canada · bluecollardev.com

- My personal web-development consultancy — custom websites, mobile + web apps, e-commerce and CRM / ERP solutions, for clients across Thailand, China, the UK and Canada. (Some open source work at github.com/bluecollardev).
- Client recommendations: Greg Lane — “an immediate increase in sales... I would definitely engage Lucas for our next project”; Brynley Cadman — contributions “innovative and valued”; Leigh Wells, P.Eng.— “never have I encountered a developer with a drive like Lucas’.”

Earlier: contracted for various tech companies in Bangkok (2006-2010); founder of one of Thailand’s early vending businesses (Siam Automated Vending Ltd. 2006–2010); taught junior-high & high-school in Bangkok (2005–2006); built my first website at age 12 (1998, still in the Internet Archive). Full role-by-role project details available on request.